

Evaluation of LLM Feature Importance Recognition and Prediction on Lung Disease

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Introduction & Motivation

Recent literature has highlighted significant sex-based performance disparities in Machine Learning (ML), both in general medical applications and within the COVID-19 domain. In cardiac disease prediction, for instance, Random Forest classifiers have been shown to overpredict disease in men (yielding higher false positive rates) and underpredict in women (resulting in higher false negative rates) when trained on sex-imbalanced datasets [1]. These disparities underscore the growing need to mitigate bias and build more equitable models.

As Large Language Models (LLMs) become increasingly integrated into sectors like Information Technology, Finance, and Medicine, concerns about fairness and representational bias have also grown. Recent studies have pointed to LLMs' tendency to favor English due to training data imbalances, leaving them underprepared for interlingual phenomena such as code-switching and code-mixing [2]. In the medical context, GPT-4 has been found to exaggerate demographic prevalence differences in 89% of diseases and generate significantly varied responses based on race and gender in clinical vignettes [3].

In this project, we compare Classical Machine Learning (CML) models and LLMs to evaluate their domain understanding and predictive performance in the context of COVID-19 and lung disease. We assess not only the prediction performance of each model when exposed to varied gender distributions in training and test datasets, but also examine the feature importance outputs, comparing the consistency and fairness of LLM-derived importance scores to those generated by CML models.

Related Works

Straw et al. quantified sex-based disparities in Random Forest cardiac-disease predictors, finding significantly higher false negative rates for women (−17.8% to −3.4%) and higher false positive rates for men (−0.5% to +9.8%) across two UCI cardiac datasets [1]. They concluded that due to the feature

importance difference caused by sex differences, there is a potential need for gender-based feature tailoring to offset these biases.

Azizi et al. trained gradient-boosting models on UK Biobank data (Mar 16–May 18, 2020). While and showed that males were more likely to test positive, older age and cardiometabolic traits predicted hospitalization in both sexes, while high-risk occupations and crowded living disproportionately increased infection risk in females [4]. However, this study does suffer from limited scope as the paper focused on predicting whether a given patient would have a positive COVID-19 test. Moreover, the age range of participants was only 40 to 60, and thus, the conclusion drawn could not be extrapolated to other age demographics.

In a Korean cohort, sex-dependent severity models saw sensitivity drops (0.93 to 0.86) and AUC decreases (0.97 to 0.94) when male-trained models were applied to female test data (and vice versa), underscoring pronounced gender bias [5]. However, it is still important to note that this dataset was relatively small, with roughly 5.6k data points. Moreover, results left off the paper—such as the 0.4063 F1 for male-dependent random forest—provide some concerns as to the overall training of the models, and if other accuracy improvements could have been made to further validate the sex-dependent sensitivity and AUC drops.

Despite this clear evidence of bias in classical pipelines, no work has systematically varied male–female training ratios nor incorporated pregnancy as a potential biasing factor. Additionally, most of the current literature focuses solely on the representing bias in Covid-19 presence, without considering other prediction columns such as symptom severity, mortality, and even development of other lung diseases like Pneumonia. The fairness of LLMs in clinical feature-importance tasks remains unexplored. Notably, GPT-4 altered clinical vignettes based on patient gender or race in 23–37% of cases, evidencing substantial demographic bias in LLM outputs [3]. Our project aimed to fill these gaps by manipulating

gender (including pregnant) distributions in both CML and LLM frameworks, comparing their feature-importance rankings and prediction metrics to uncover potential biases.

Dataset & Exploratory Data Analysis

Data Distribution & Cleaning

The dataset we selected was provided by *La Secretaría de Salud Federal de México* (Mexican Federal Secretary of Health), which provided an enormous dataset of 1,048,576 unique, anonymized patient data. This was a key factor as previous studies that attempted to compare LLMs to CML in the domain of COVID-19 for mortality prediction did not have access to this many data points, and thus relied on oversampling techniques like SMOTE to increase the amount of available training data [6]. The dataset was comprised of the following feature categories:

- General Patient Information
- Hospital Care Received
- Respiratory Conditions & Related Factors
- Chronic & Pre-existing Conditions
- Pregnancy Status
- Severe Case/Symptom Indicators

In total, the dataset consisted of 18 feature columns which we utilized for training, and 3 output columns for prediction.

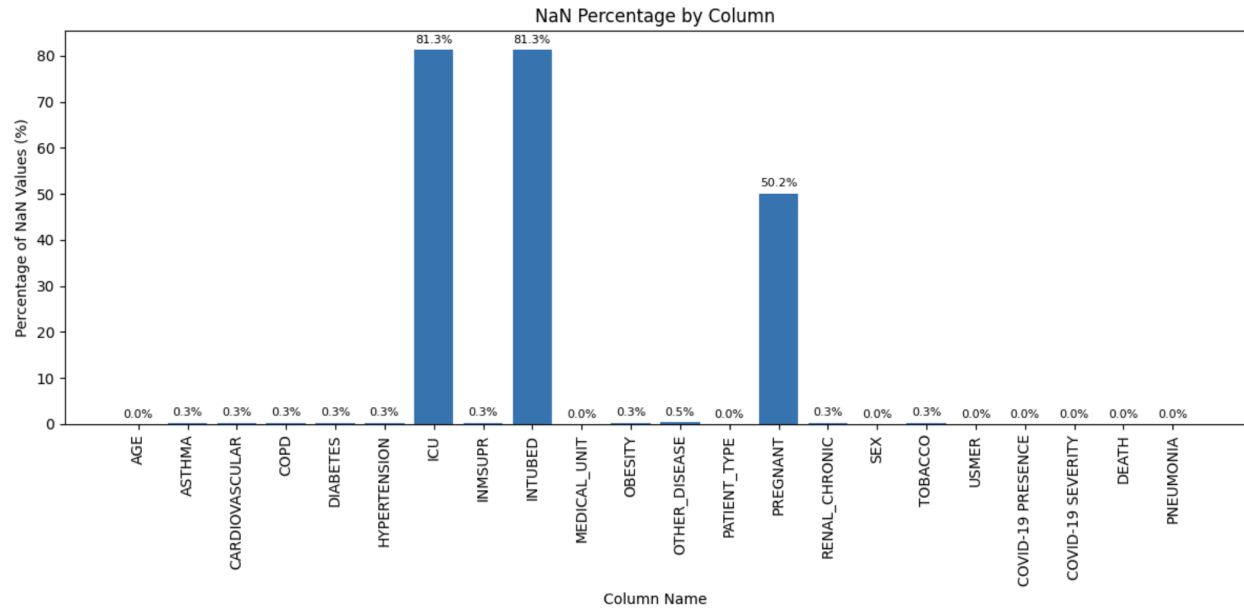


Figure 1: NAN Percentage by Column

We also wanted to analyze the percentage of NaN values present in our clinical data. As Tan et al. showed in their paper, higher patient loads during COVID-19 surges had systematically more missing labs, suggesting that overwhelmed staff and resource constraints directly led to gaps in recorded data [7]. While it is not uncommon for medical data to contain missing values due to record keeping errors, we wanted to gain a better understanding of how this impacted our selected dataset. As shown above, the ICU and Intubed contained the highest NaN value rates at 81.3%, however, when looking into the dataset description, this was expected ICU and Intubed was only reported if a patient was admitted to the ICU unit. The other key column we see was Pregnancy, however, this column remains unreported for males.

Distribution of Prediction Columns

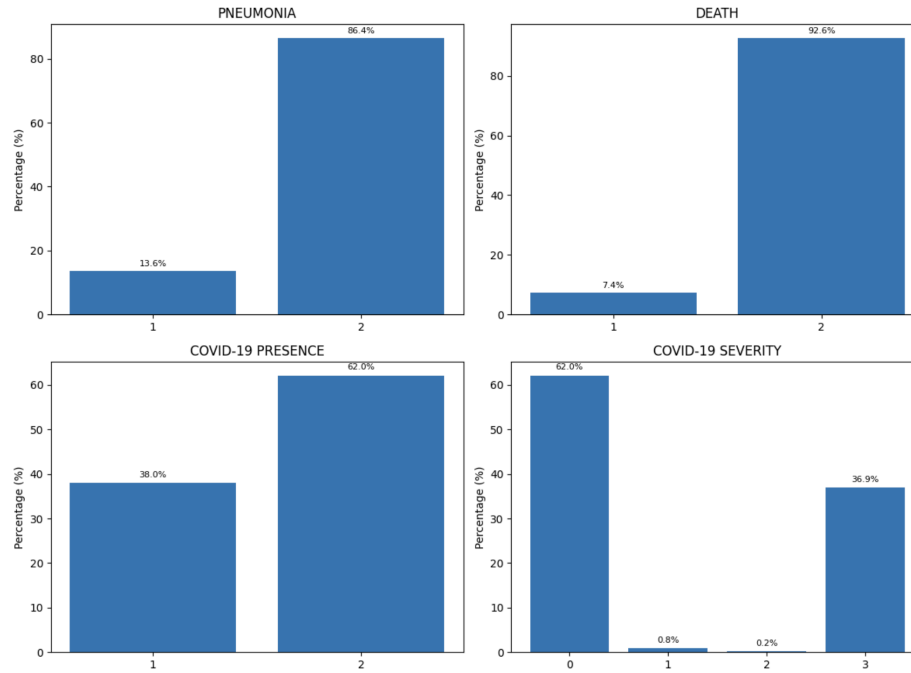


Figure 2. Distribution of Prediction Columns

Prediction Column Distributions by Gender Group

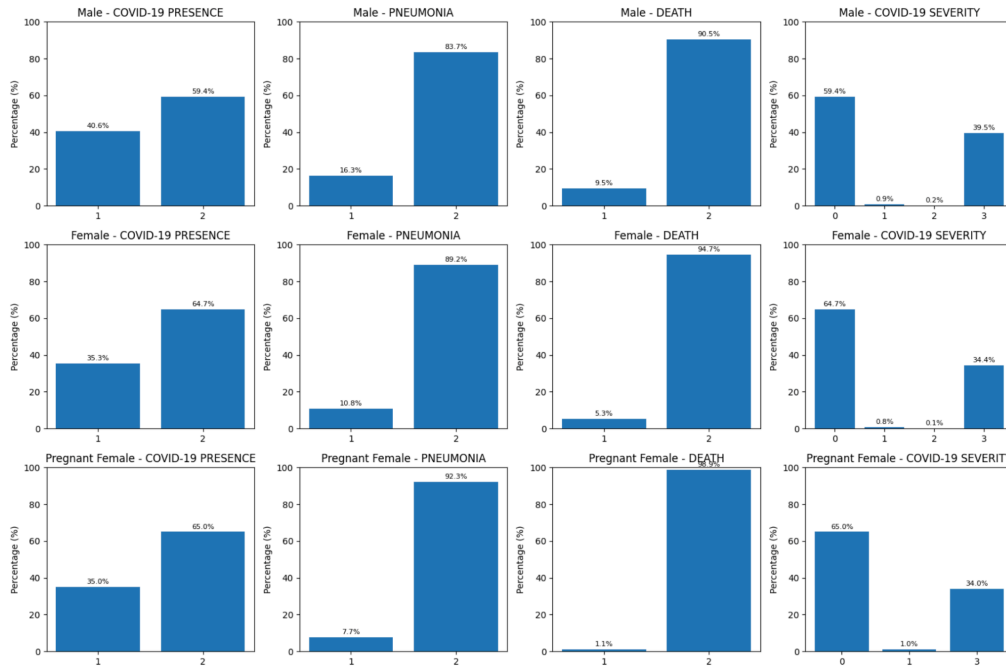


Figure 3: Prediction Column Distribution Across Various Gender Distributions

We initially assumed that our data would have an even distribution across the prediction columns, however, when we plotted the distributions, we uncovered that this was not the case. We note the very high skew for Pneumonia and Death, as there was an extremely low percentage for “yes” (class label 1). The uncleaned data also only contained the COVID-19 Severity Column, which represented no COVID-19 presence (class 0) to the symptom severity (1 being low, 3 being high). To provide more balancing, we engineered a new output prediction column: COVID-19 Presence. This simply remapped the severity scores into a binary classification setting, providing a more balanced distribution and adding a new output column. With all this in mind, we recognized that we would have to use balancing techniques for the CML training in order to mitigate training bias caused by data skewness (see Class Weighting section).

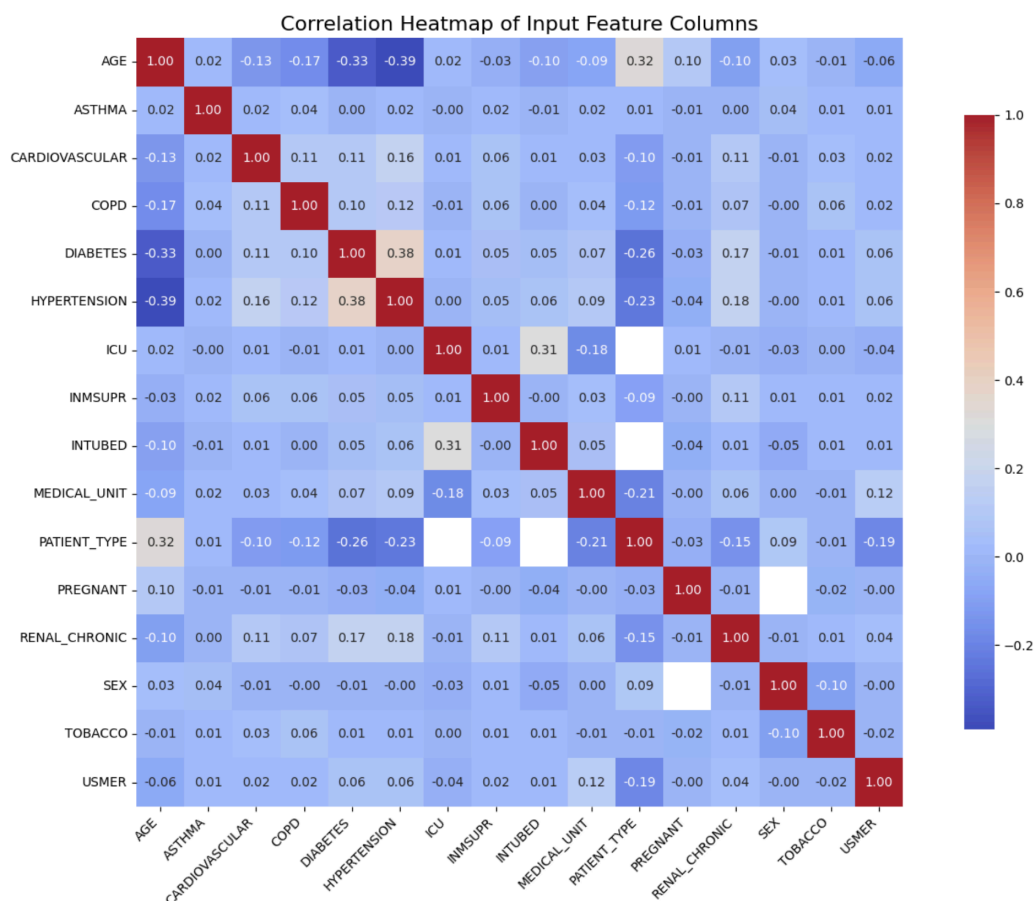


Figure 4: Heatmap of Input Feature Columns

Conversion from Tabular Data to Prompt-Based Dataset

This process was assisted by the usage of GPT-4o mini to provide a script that would handle this conversion process automatically. This was done to ensure we did not introduce random noise or perturbations in the data we fed the LLMs by providing a single, reusable structure, as the tabular data being seen by our CML models is not subject to such changes.

Our first attempt was rather naive, as we simply prompted GPT-4o mini to convert single rows of clinical data into a prompt simply listing of “Feature: Value” pairs. However, isolated tokens like “Tobacco: 2” lacked narrative coherence for an LLM. Inspired by the reviewed literature’s emphasis on clarity, we next asked GPT-4o mini itself to convert data from “Feature: Value” pairs into complete sentences. The prompts then began with “The patient is a 58.0-year-old female categorized as patient type 2.0...,” stringing together each binary and numerical attribute in a flowing paragraph. To further improve prompt coherence, structuring, and alignment with traditional english speaking, we utilized *Large Language Models versus Classical Machine Learning: Performance in COVID-19 Mortality Prediction Using High-Dimensional Tabular Data Section 2.4.4 Preparing Data for LLM* as a basis for our conversion of tabular data to a textual, prompt-based dataset [8]. From this numeric features were embedded in full-sentence form for readability. For age, we prefaced the value with “the patient’s age is” (e.g., “the patient’s age is 58.0 years old”). Sex was encoded simply as “male” or “female,” reflecting common demographic descriptors in clinical patient descriptions.

While we had included missing or “unknown” values—e.g., “has an unknown ICU admission status”—in our earlier interactions, Professor Pu cautioned that providing unknown flags gives the LLM side information our classical models wouldn’t see. Thus in our final prompt design, we decided to omit any mention of NaNs, leaving out those attributes entirely. This also aligned more with common speech, as unknown features are typically omitted when describing a patient, rather than listing all unknowns. This

ensures our prompt-based LLM and our CML models (CatBoost and XGBoost) baselines work from equivalent information, making comparisons fair and preventing inadvertent prompt leakage.

Standardization of Data Splitting

To ensure a fair comparison between our LLM and classical machine learning (CML) models, we needed both approaches to evaluate the same testing sets. We used the `train_test_split` function from `scikit-learn`, setting a fixed random seed. This allowed us to consistently sample the same data points across different gender distributions (e.g., all male, all female, 50-50 split), ensuring alignment between the tabular and prompt-based datasets without any variation across runs.

Implementation & Methodology

LLM Model Selection

Our chosen large language models: ChatGPT-4o (OpenAI), Claude 3.7 Sonnet (Anthropic), and DeepSeek Deepthink R1 were selected based on the claims of strengths made by their respective companies. Given our priority was to evaluate the medical attribute significance across diverse patient datasets and our chosen target columns, we considered their ability to reason, transparency in their shortcomings, and complex architecture.

1. ChatGPT-4o (“omni”)

OpenAI’s LLM was chosen for its capability of multimodal reasoning and its position as the most rapid, technologically advanced, and inexpensive model since its May 2024 release [13].

- a. Claim: Especially strong at logic tasks, structured understanding, and judgement in relevance of text data. It offers great performance related to math, reasoning, and consistency of factual information.

- b. Connection: As we fed it our features to rank, since it was devoid of actual data, we relied on its semantic comprehension and medical application from its domain information. It has shown adequate performance in benchmarks such as MMLU and MedQA.

2. Claude 3.7 Sonnet (Anthropic)

Anthropic's Claude model was handpicked for its prioritization of safety, strong reasoning abilities, and affirmed performance in the sector of data science. Generally, Claude LLMs rank exceptionally high on benchmarks code reasoning (HumanEval), GPQA, and MMLU (similar to ChatGPT-4o), where it places careful importance in logical ordering [14].

- a. Claim: Excels at interpreting obscure prompts that examine human intent thereby producing useful, structured outputs. It was critically trained to reduce harmful bias and provide fairness. It contains an internal ranking system and confidence scoring in specified tasks.
- b. Connection: Since models receive only a list of clinical features (without any of our raw data), it is integral to remain context sensitive when analyzing gendered and pregnancy-related subgroups. We knew that unconscious bias in AI models could influence our ranking system.

3. DeepSeek Deepthink R1

DeepSeek AI was hailed as a pillar of developing, open-source large learning models trained with an emphasis on reasoning and scientific information. DeepSeek is novel in its transparency and domain-specific strengths (math, code, and scientific problem-solving) [12].

- a. Claim: According to Deepseek's benchmark reports, it details incredible performance in structured reasoning and data interpretation differentiating them from standard

proprietary LLMs. It places specific assertions on its superiority in context reasoning and cognizance prompts [11].

- b. Connection: As it relates to our feature importance ranking, we evaluated its comparative abilities to ChatGPT-4o and Claude from its open-source nature to order our relevant medical features despite the lack of real data and no prior manual fine-tuning.

Each was chosen based on their reputation, varying training philosophies, and interpretability in the LLM industry. Our intention was to see not only their accuracy and explainability of such rankings resulting in phenomenal performance, but to probe their interpretations on medical factors in the context of real patient data.

Prompt Engineering for Feature Importance Recognition

Over the course of three iterations (see Figures 1–3), we honed our zero-shot feature-importance prompt to rely solely on LLM domain knowledge and to mirror the “weight” metric used by tree-based models. We chose to focus on weight rather than gain or cover for measuring feature importance, both for conceptual alignment and fairness. In XGBoost and CatBoost, weight is defined as the number of times a feature is used in a split across all trees in the ensemble, reflecting how often the model relied on that feature when constructing its internal decision logic. On a high level, this shows “what the model paid attention to” based on the structure of the trees, not necessarily what had the most impact on the final prediction. On the other hand, gain measures the actual improvement in the optimization criterion each time a feature is used. Similarly, cover reflects the number of data points affected by each split. Since our LLMs are not seeing actual data, and are instead operating entirely based on domain knowledge and prior training, *gain* and *cover* would unfairly skew the comparison as they both tie feature importance to the actual improvements made from the training data .

In attempt 1, our prompt asked the model to rank 18 features “based on feature importance weight analysis as used with random forest/tree algorithms by XGBoost and CatBoost,” and referenced our

dataset. We assumed this would provide contextualization, however, as shown in the figure below, the model repeatedly requested the actual CSV file and fixated on dataset provenance—prompting us to strip out any file mentions in future prompt interactions.

```
Let's start by analyzing the dataset and calculating feature importances using XGBoost and CatBoost.
Please upload the CSV file so I can proceed.
```

Figure 5: ChatGPT Requesting CSV File when Extracting Feature Importance

In attempt 2, we removed the CSV reference and dataset source, clarified that outputs must be JSON key–value pairs, and emphasized a 0–1 scale. This was done in part to standardize our weighting of feature importance. Since weight measures how many times a feature is used in a split across all trees in the ensemble, we could get drastically different numerical outputs as the LLM makes assumptions about tree depth, pruning, and other hyperparameters. Thus, with a 0-1 scale, this could make the outputs more interpretable while still getting a semblance of feature weighing. However, Deepseek assigned raw scores but failed to normalize them to sum to 1, while only ChatGPT attempted normalization—ending at 1.05 instead of exactly 1. This taught us to explicitly require normalization.

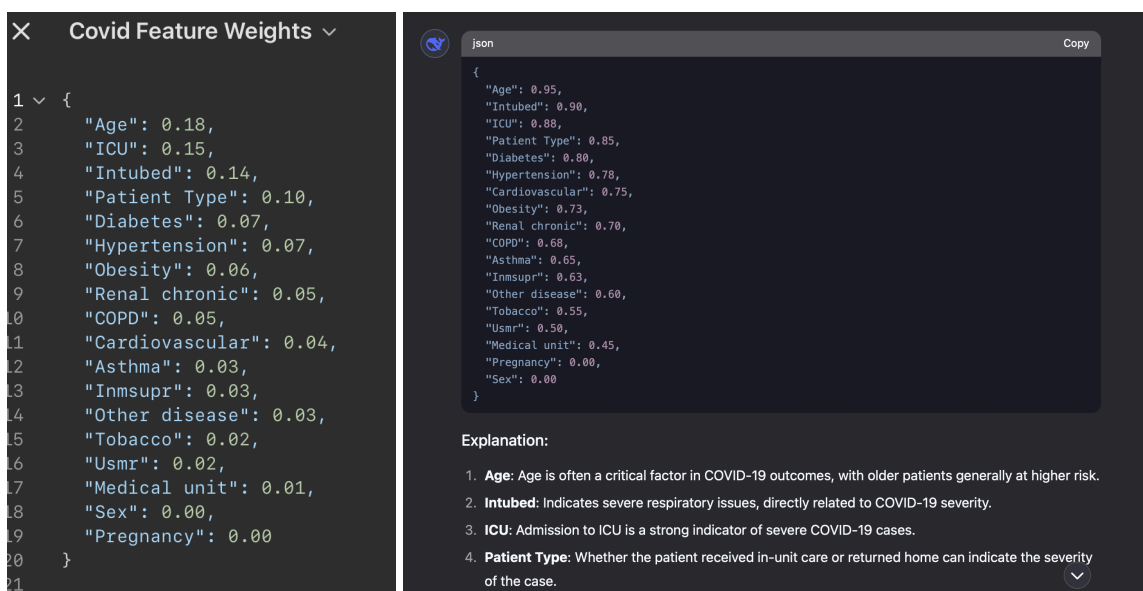


Figure 6: ChatGPT and DeepSeek Variations in Feature Importance Normalization

In attempt 3, we ran the refined prompt across three LLMs (ChatGPT-4o, Claude 3.7 Sonnet, Deepseek), two chain-of-thought settings (CoTR enabled/disabled), and five gender splits. By appending “Ensure the sum of all feature-importance weights equals 1.0,” every model finally respected both format and scale. See Appendix for finalized feature importance prompt.

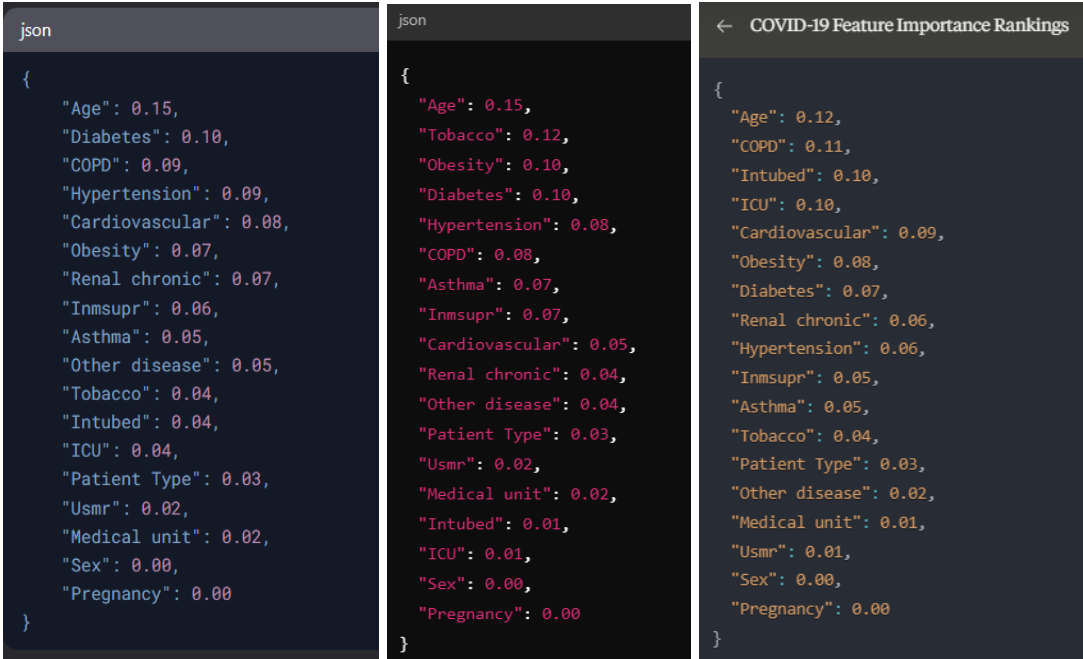


Figure 7: Example of Properly Structured Feature Importance from selected LLMs

Classical Machine Learning Training Script and Metric Collection

CatBoost Overview

CatBoost is a gradient-boosted decision-tree ensemble specifically optimized for handling categorical features without explicit encoding or preprocessing [9]. It builds trees sequentially, where each new tree corrects errors from the previous ensemble via gradient boosting. Importantly, CatBoost can internally process missing values by learning optimal split directions for NaNs during training. This helps to preserve real data sparsity and avoid potentially bias-inducing imputation in medical datasets such as the data we are using [9].

XGBoost Overview

XGBoost (“Extreme Gradient Boosting”) is also a gradient-boosted tree ensemble but extends the classic framework with explicit regularization (L_1/L_2 penalties) to prevent overfitting. Additionally, it also handles missing values by assigning them default split directions in their algorithm, making it sparsity-aware [10]. By leveraging both the gradient and Hessian of the loss function, XGBoost makes more informed split decisions and scales efficiently to large, high-dimensional datasets, all while still having native support for NaNs values [10].

Training Script

Our training pipeline begins by computing class weights from the target-class frequencies. This counteracts severe skew in outcomes—mentioned early in exploratory data analysis section—by passing balanced weights into the booster’s `scale_pos_weight` (XGBoost) or `class_weights` (CatBoost) parameter. We then perform a grid search over hyperparameters, wrapping each candidate configuration in 5-fold cross-validation on the training split. For each fold, the script fits the model with the computed class weights, evaluates performance metrics, F1 score for binary classification and Total F1 score of multi-class classification, and records the average score. After performing the grid search, we select the hyperparameter set with the best mean validation score, retrain on the full training data, and finally evaluate on the held-out test set.

We ran this script for all combinations of gender distribution (All Male, All Female, All Pregnant Female, Original Unbalanced Data, 50-50 Gender Balanced) and all output columns (COVID-19 Severity, COVID-19 Severity, Death, Pneumonia), collecting the normalized feature importance based on weight, and the performance metrics (Accuracy, Precision, Recall, F1, etc).

Prompt Engineering for LLM Metric Collection

To evaluate the predictive performance of large language models (LLMs) on structured clinical data, we implemented a two-step pipeline aligned with our classical machine learning (CML) experiments.

Data Preparation and Prompt Structuring

The first stage, handled by our LLM Prediction Split notebook, involved creating standardized CSV files that matched the exact test splits used in the CML pipeline. These splits were generated by joining tabular clinical features with their corresponding prompt-formatted patient descriptions. We stratified the data across five gender-based distributions—All Male, All Female, All Female Pregnant, Gender Balanced (50/50), and Gender Unbalanced—and four prediction targets: COVID-19 Presence, COVID-19 Severity, Death, and Pneumonia. To manage prompt size limits in ChatGPT and prevent browser instability, we sampled up to 100,000 rows per gender-target split, preserving the class distributions through random sampling. Each split was saved in two formats: one version containing ground truth labels (used for post-prediction metric comparison), and one with the target column masked (used as input for LLM prompting).

ChatGPT was then prompted with the masked version of each dataset using a standardized prompt format: a zero-shot instruction that described the task (e.g., “predict COVID-19 Presence”), included a glossary of feature columns and definitions, specified the gender distribution, and defined the expected output format. The model responded with completed CSVs where the target column was populated based on its inferred prediction logic.

Metric Evaluation

The second stage of the workflow, conducted in the LLM vs. CML Metric Comparison notebook, evaluated the predictive performance of ChatGPT's outputs using the same metrics used for XGBoost and CatBoost: accuracy, macro and weighted F1 scores, precision, recall, and ROC AUC (where applicable). For consistency, the notebook aligned LLM-predicted rows with the ground truth data based on order and sample count. Custom metric functions handled both binary and multiclass cases (e.g., 4-class COVID-19Severity), while confusion matrices and class distribution differences were also calculated. These results were then stored in JSON format for each gender distribution and target value, and plotted for visual comparison alongside CML model metrics across all gender splits and target variables.

This methodology allowed for a fair, side-by-side comparison of ChatGPT's ability to generalize from patient feature descriptions to clinical outcome prediction, benchmarked against robust tree-based classifiers trained on the same data.

```
{
  "accuracy": 0.6432521358441725,
  "f1_macro": 0.46590579760798334,
  "f1_weighted": 0.5563831629336142,
  "confusion_matrix": [
    [
      312817,
      19108
    ],
    [
      163914,
      17190
    ]
  ],
  "label_distribution_comparison": {
    "0": {
      "actual_count": 331925,
      "predicted_count": 476731,
      "difference": 144806
    },
    "1": {
      "actual_count": 181104,
      "predicted_count": 36298,
      "difference": 144806
    }
  },
  "total_actual": 513029,
  "total_predicted": 513029,
  "dataset": "Test",
  "model_name": "Chatgpt"
}
```

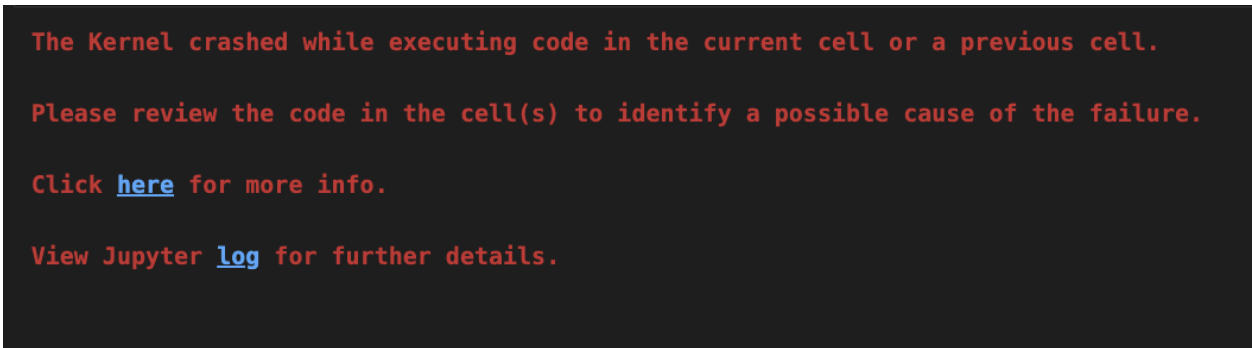
Figure 8: Example of Properly Structured Chatgpt Metric Output File

Major Blockers

/Users/KailenMcCauley/anaconda3/envs/VIP/lib/python3.11/site-packages/joblib/externals/loky/process_executor.py:752: UserWarning: A worker stopped while some jobs were given to the executor. This can be caused by a too short worker timeout or by a memory leak.

Figure 9: Memory Leak Error

A major blocker was caused due to a Kernel crash while running the CML training script. Both models could be trained, but when attempting to calculate their respective metrics from the predictions and true labels, the following error would occur.



The Kernel crashed while executing code in the current cell or a previous cell.
Please review the code in the cell(s) to identify a possible cause of the failure.
Click [here](#) for more info.
View Jupyter [log](#) for further details.

Figure 10: Kernel Crash for Jupyter Notebook while Calculating Metrics

This was extremely difficult to debug, as the error itself did not produce a stack trace, meaning that we had to manually go line by line through the code to resolve the issue. Initially, our tracing led us to the calculation of true positive, false positive, true negative and false negative when it was a multi-class classification. After testing this logic, it became clear this was not in fact the root of our issue. Continuing to trace backwards, we discovered a dimensionality issue between our predicted and true tensor. Our predicted tensor was a matrix of 1 by n, where n is the number of datapoints in our test set. However, our true values were only an array of size n. This shape difference leads to numpy implicit broadcasting, making a massive temporary object and crashing memory. After resolving this by converting the predicted tensor matrix to an array, the error was resolved.

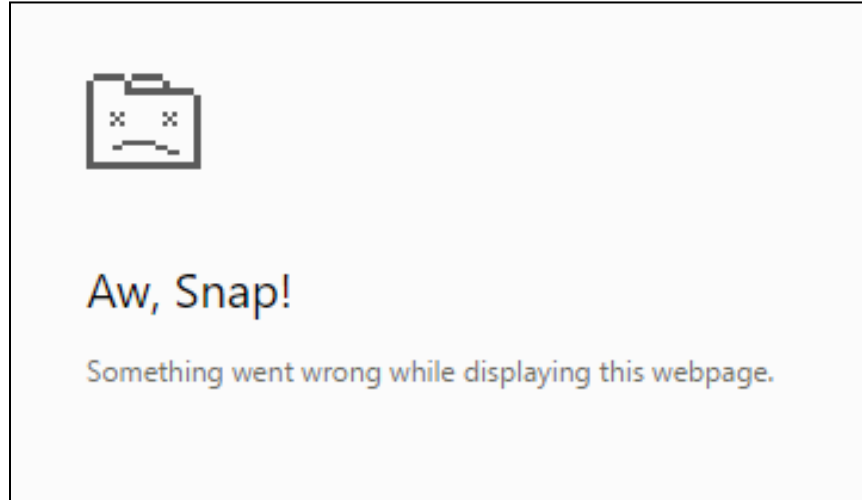


Figure 11: Browser crash error caused by ChatGPT attempting to process large CSV input files.

Another major issue throughout this project was browser instability when prompting ChatGPT with large datasets, especially for the GenderBalanced and GenderUnbalanced splits. Attempting to upload full datasets often led to crashes (see **Figure 11**), forcing us to cap each dataset at 100,000 rows to prevent interruptions.

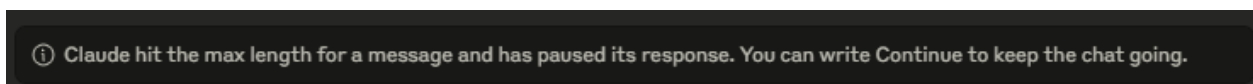


Figure 12: Claude max conversation length warning.

We also attempted to run the same tasks using Claude 3.7 Sonnet and Deepseek, but both platforms had a hard upload limit of ~65,000 rows and failed to execute predictions reliably due to prompting size restrictions. Additionally, these models returned incompatible outputs, even when explicitly instructed to produce CSVs. As a result, we relied solely on ChatGPT for LLM metric generation, with dataset size limits in place to ensure stability.

Results

CML Results & Analysis

Feature Importance

Analysis of the feature importance values for both XGBoost (Figure 14) and CatBoost (Figure 13) reveals interesting patterns in how these classical ML models prioritize features when predicting COVID-19 outcomes. Both models consistently rank Age as the most important predictor, with XGBoost assigning it 0.32 average importance and CatBoost giving it an even higher 0.35 average importance. This strong agreement between the models reinforces clinical understanding of age as a critical risk factor for COVID-19 severity.

However, the models diverge significantly in their secondary feature priorities. XGBoost identifies Medical unit (0.14) as the second most important feature, while CatBoost assigns much lower importance to this feature (0.13). Instead, CatBoost appears to place greater emphasis on Patient Type (0.06) compared to XGBoost (0.04).

For comorbidities, CatBoost assigns higher importance to several conditions compared to XGBoost. For instance, CatBoost values COPD at 0.03, Cardiovascular at 0.05, and Diabetes at 0.04, while XGBoost assigns 0.023, 0.027, and 0.038 respectively. This suggests CatBoost may be more sensitive to certain clinical risk factors in its decision-making process.

Notably, both models assign relatively low importance to gender-specific features, with XGBoost rating Sex at just 0.004 and Pregnancy at 0.025, while CatBoost values these at approximately 0.004 and 0 respectively. This similarity indicates that despite differences in their algorithmic approaches, both tree-based models found minimal predictive value in gender information compared to clinical indicators and institutional factors.

The consistency in ranking Age as the dominant predictor while showing algorithmic-specific variations in secondary feature importance demonstrates how different CML approaches can extract distinct patterns from the same underlying data.

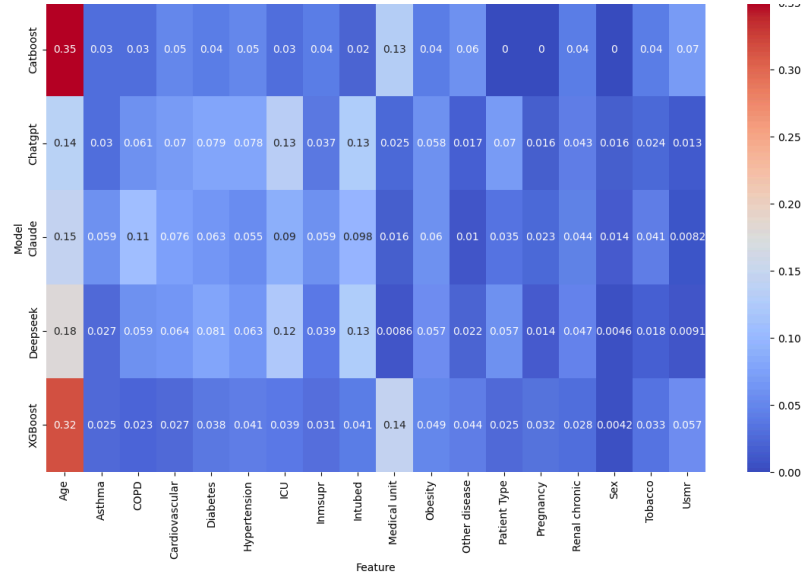


Figure 13: Heatmap of Average Feature Importance: CatBoost

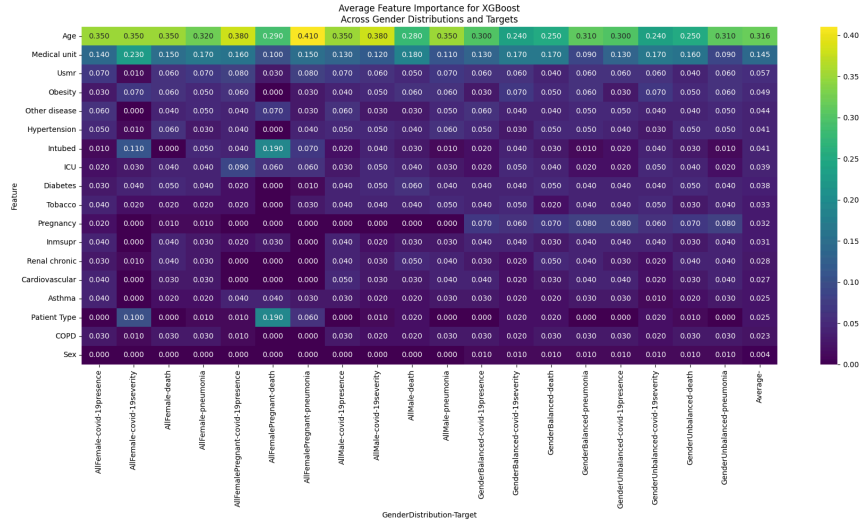


Figure 14: Heatmap of Average Feature Importance Across Gender Distributions: XGBoost

Data Prediction: CatBoost

For COVID-19 presence prediction, CatBoost achieves consistent performance with accuracy ranging from 64.5% (AllMale) to 67.9% (AllFemalePregnant). Notably, the model demonstrates higher recall (85-86%) than specificity (28-35%) across general populations, indicating a tendency to identify positive cases more reliably than ruling out negative ones. The AllFemalePregnant subset shows better balance between recall and specificity (66.6% and 70.3% respectively), suggesting more equitable performance for pregnant populations.

For death prediction, CatBoost achieves impressive performance with accuracy exceeding 87% across all gender distributions. The model shows exceptional precision (>99%) but somewhat lower recall (86-90%), indicating it rarely gives false positive death predictions but sometimes misses true positive cases. Performance is remarkably consistent between gender distributions, with minimal variation in F1 scores (93.5-94.8%).

For pneumonia prediction, performance is similarly strong with accuracy around 89-90% across gender distributions. The model maintains high precision (>97%) with slightly higher recall (89-91%) than for death prediction. Again, gender-based variations are minimal, suggesting the model generalizes well across demographic groups for this outcome.

For COVID-19 severity (multiclass), the model struggles significantly as shown in its confusion matrices (Figure 15), with accuracy dropping to 42-50% across distributions. Class-specific analysis reveals the model performs well on the predominant classes (0 and 3) but struggles with minority classes (1 and 2), achieving F1 scores of only 25-29%. This indicates challenges in distinguishing intermediate severity levels.

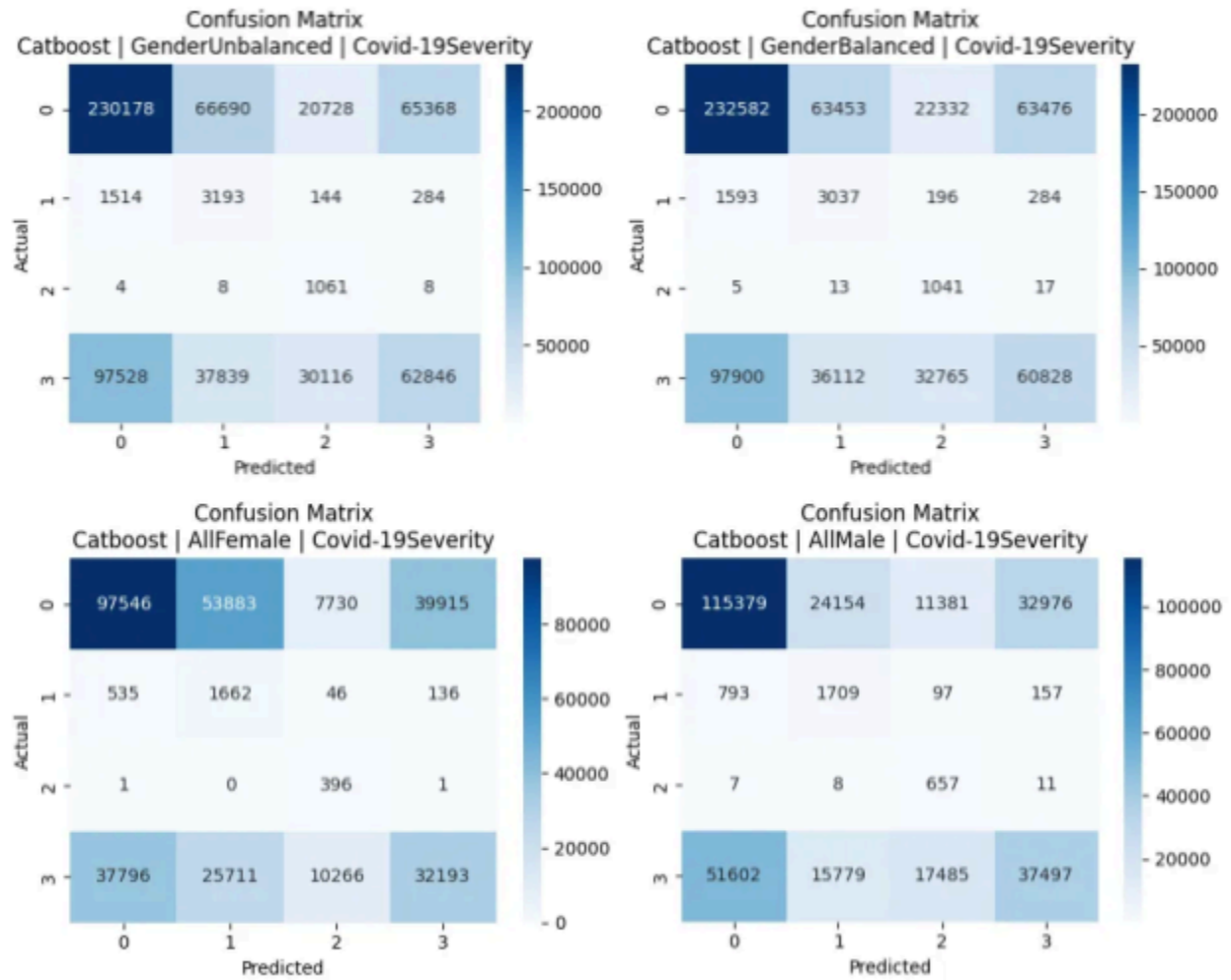


Figure 15: Distinctly low accuracy, precision, and recall as demonstrated by Covid-19 Severity confusion matrices with CatBoost training


```

1  {
2      "tp": 170809,
3      "tn": 31214,
4      "fp": 77229,
5      "fn": 28565,
6      "specificity": 0.28783785029923553,
7      "accuracy": 0.6563087808665539,
8      "precision": 0.6886404502535902,
9      "recall": 0.856726554114378,
10     "f1": 0.7635423278767668,
11     "f2": 0.8168505280555199,
12     "roc_auc": 0.605551068966278,
13     "au_prc": 0.7201964175513904,
14     "sensitivity": 0.856726554114378,
15     "confusion_matrix": [
16         [
17             31214,
18             77229
19         ],
20         [
21             28565,
22             170809
23         ]
24     ],
25     "dataset": "Test",
26     "model_name": "CatBoost"
27 }

```

Figure 16: CatBoost Metric JSON file format for AllFemale gender distribution, target value: Covid-19 Presence

Comparing gender distributions, the Gender Balanced and Gender Unbalanced groups show nearly identical metrics across all prediction tasks, suggesting the model's gender composition sensitivity is minimal for established conditions. However, the AllFemalePregnant distribution predictions show distinct performance patterns, particularly for COVID-19 presence prediction, where the model achieves better specificity than with other populations.

These findings demonstrate that while CatBoost exhibits robust performance for binary outcomes (particularly death and pneumonia), it struggles with multi-class prediction and shows some sensitivity to pregnant populations in certain prediction tasks.

Data Prediction: XGBoost

For COVID-19 presence prediction, XGBoost demonstrates high recall but low specificity across all gender distributions. The model achieves accuracy between 65-67%, with particularly high recall rates (90-99%) across populations, indicating a strong tendency to classify cases as positive. The AllFemalePregnant distribution shows the most extreme imbalance, with recall of 99.7% but specificity of only 6.1%, suggesting the model overwhelmingly predicts COVID-19 presence in pregnant populations.

For death prediction, XGBoost achieves impressive accuracy (93-95%) with extremely high recall (98-99%) across all gender distributions. However, specificity varies considerably, from as low as 21.4% in AllFemalePregnant to 46.2% in AllMale populations. This indicates that while the model rarely misses positive death cases, it frequently classifies non-death cases as deaths, particularly in pregnant populations.

For pneumonia prediction, performance is similarly strong with accuracy around 90-93%. The model maintains high recall (96-99%) but lower specificity (18-60%) across gender distributions. The AllFemalePregnant distribution again shows the most extreme imbalance with recall of 99.7% but specificity of 18.5%.

For COVID-19 severity (multiclass), XGBoost exhibits a striking pattern across all gender distributions: it only makes predictions for classes 0 and 3, completely ignoring intermediate severity classes (1 and 2) (Figure 17). This represents a significant limitation in the model's ability to discriminate between different severity levels. Despite overall accuracy of 65-67%, the F1 scores are very low (0.27-0.34) due to this class imbalance.

Comparing performance across gender distributions reveals consistent patterns: XGBoost generally achieves the highest F1 scores in the AllMale population, while showing the most extreme

recall-specificity imbalances in the AllFemalePregnant data. The Gender Balanced and Gender Unbalanced distributions show nearly identical performance metrics across all prediction tasks.

When comparing XGBoost to CatBoost, XGBoost consistently shows higher recall but lower specificity, indicating a stronger bias toward positive predictions. While this results in fewer missed cases, it also leads to more false positives. Additionally, XGBoost's binary approach to COVID-19 severity prediction (only using classes 0 and 3) contrasts with CatBoost's attempts to model all severity classes, suggesting fundamental differences in how these algorithms handle multiclass prediction with imbalanced data.

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2      "tp": 192939,
3      "tn": 13971,
4      "fp": 94472,
5      "fn": 6435,
6      "specificity": 0.12883265863172358,
7      "accuracy": 0.6721850969894451,
8      "precision": 0.6712999850388468,
9      "recall": 0.9677239760450209,
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13     "au_prc": 0.7382322638902376,
14     "sensitivity": 0.9677239760450209,
15     "confusion_matrix": [
16         [
17             13971,
18             94472
19         ],
20         [
21             6435,
22             192939
23         ]
24     ],
25     "dataset": "Test",
26     "model_name": "XGBoost"
27 }
```

Figure 17: XGBoost metric JSON file format for All Female gender distribution, target value: Covid-19 Presence

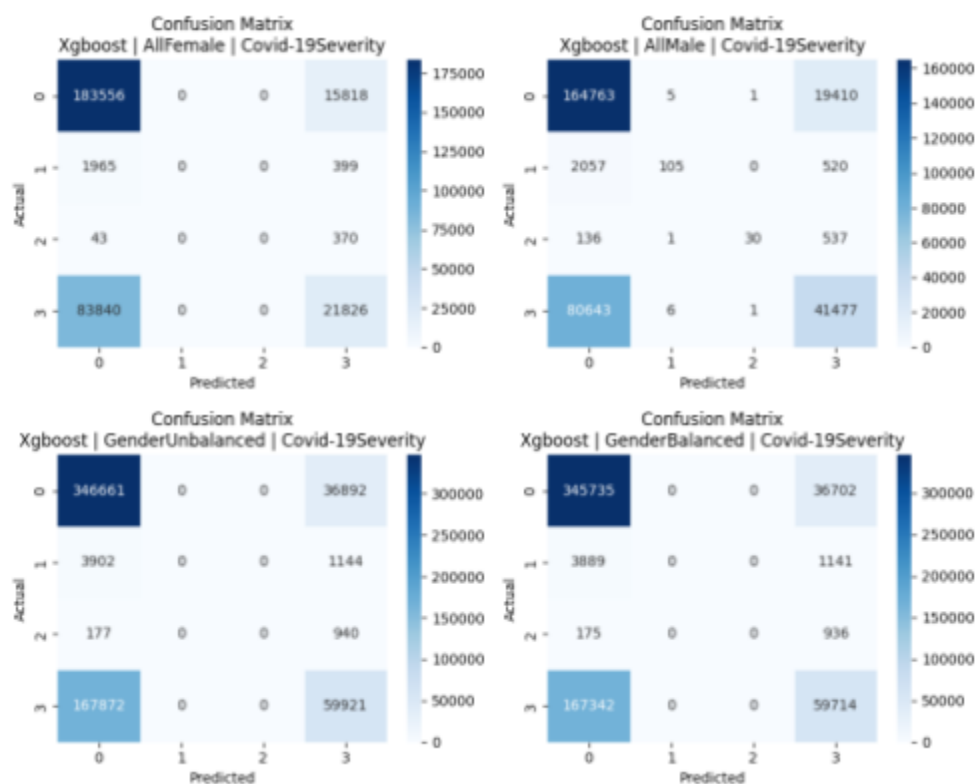


Figure 18: XGBoost demonstrates higher class imbalance and lower accuracy in multi-class predictions for Covid-19 Severity.

LLM Results & Analysis

Feature Importance

Our comprehensive analysis of feature importance visualization across multiple LLMs revealed consistent and meaningful patterns in how these models prioritize clinical features for COVID-19 and lung disease prediction. The heatmap visualization (Figure 19) demonstrates remarkable consistency in feature ranking across gender distributions and target variables, with Age (0.141), Intubed status (0.112), and ICU admission (0.111) emerging as the most significant predictors, followed by COPD (0.084) and Cardiovascular conditions (0.073).

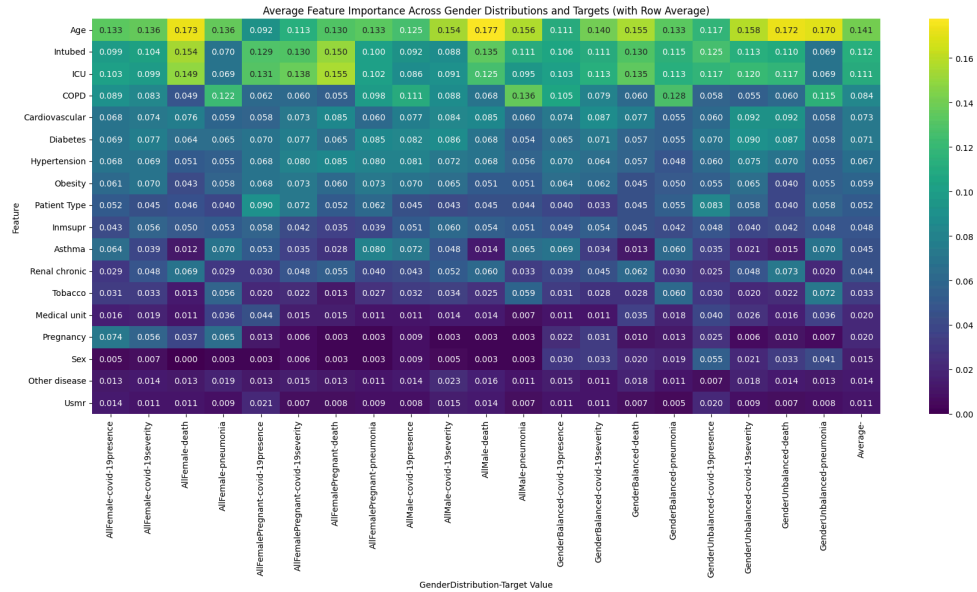


Figure 19: Average Feature Importance Across Gender Distributions and Target

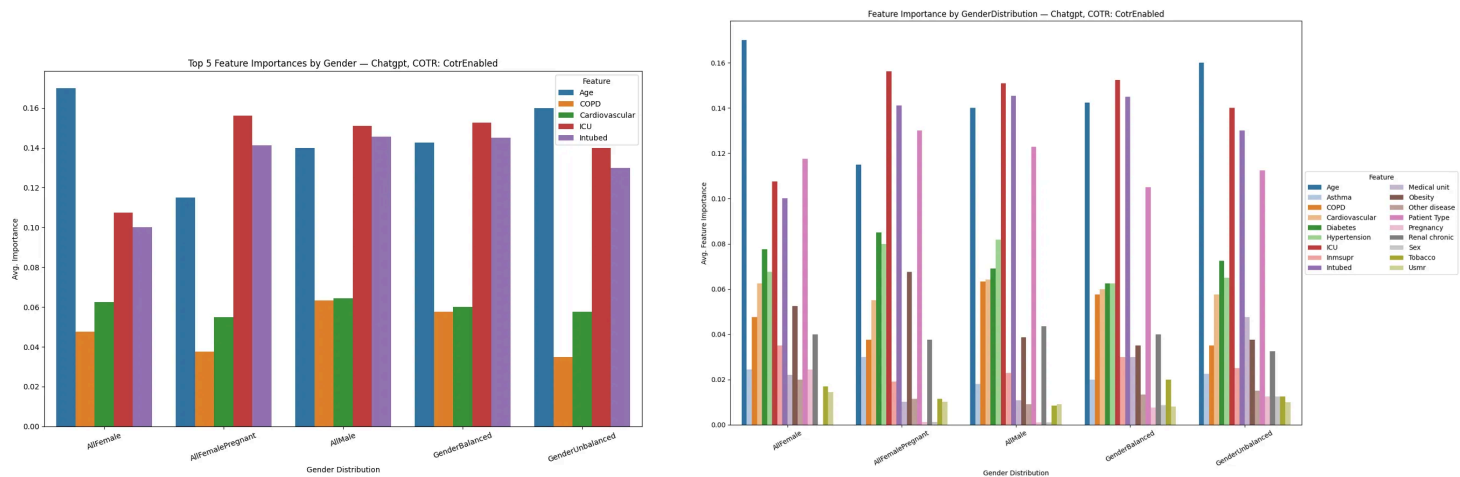
Chain-of-Thought Effects on Feature Importance in ChatGPT

Comparing ChatGPT's CoTR-enabled (Figure 20 - 21) versus CoTR-disabled (Figure 22-23) visualizations reveals significant differences in feature importance distribution across gender distributions. With CoTR disabled, ChatGPT assigned extremely high importance to Age for AllMale (0.172) while showing more balanced feature distribution for female populations. In contrast, CoTR-enabled ChatGPT demonstrated more nuanced gender-specific feature prioritization, particularly for pregnant populations.

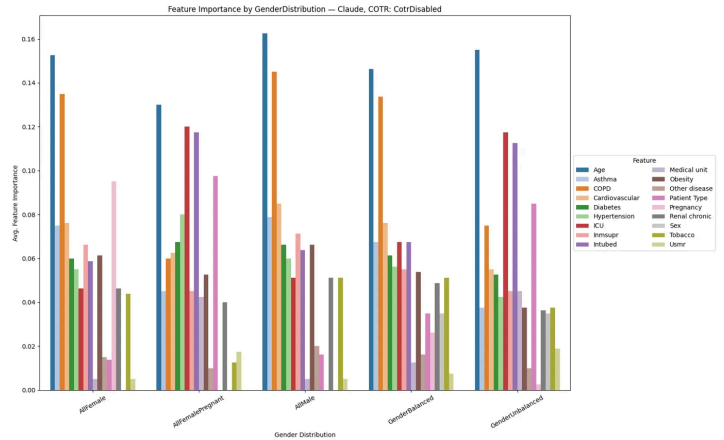
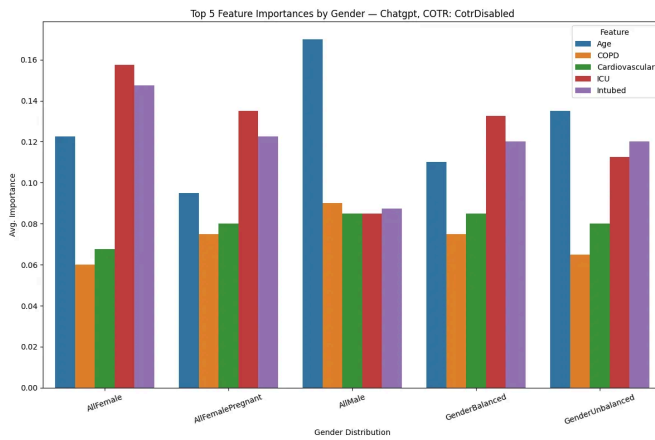
One striking difference appears in pregnancy-related features. CoTR-enabled ChatGPT (Figures 14, 15) assigned substantially higher importance to Pregnancy status in the AllFemalePregnant group (0.130, appearing as a prominent pink bar) compared to the CoT-disabled model (Image 3) where Pregnancy

importance remained minimal (0.024). This suggests that explicit reasoning pathways enable ChatGPT to better incorporate pregnancy as a clinically relevant factor when appropriate.

Furthermore, CoTR-enabled ChatGPT showed more balanced feature importance across gender groups, particularly for comorbidities. For example, Cardiovascular importance values became more consistent across gender distributions (0.055-0.065) when CoTR was enabled, compared to the more variable distribution (0.058-0.084) in the CoTR-disabled model. Lastly, ICU and Intubed features notably differ in importance between AllMale and AllFemale when CoTR is enabled versus disabled. Enabling CoTR seems to increase the feature importance of Age in AllFemale gender distribution across all target values, while decreasing ICU, and increasing the feature importance of ICU and Intubed in male populations, whereas disabling CoTR has the exact opposite affect – drastically increasing feature importance of ICU and Intubed on AllFemale populations while dramatically decreasing those features on AllMale data.



Figures 20, 21: Top 5 Features and Feature Importance by Gender, Chatgpt, CoTREnabled

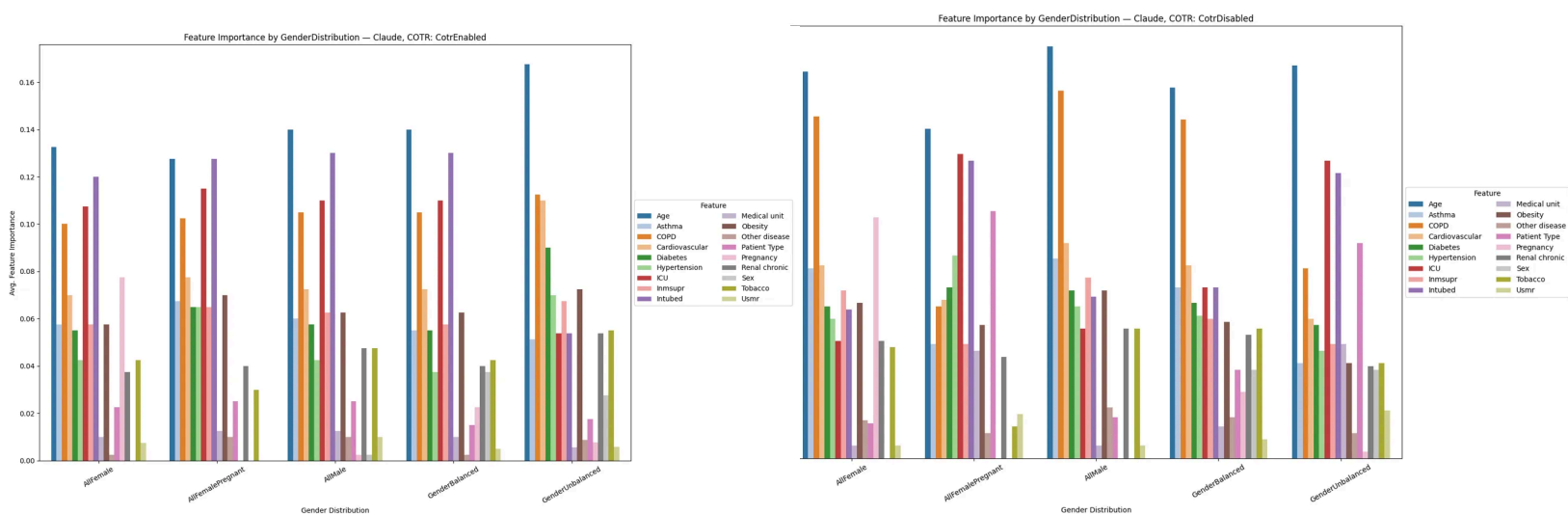


Figures 22, 23: Top 5 Features and Feature Importance by Gender, Chatgpt, CoTRDisabled

Chain-of-Thought Effects on Feature Importance in Claude

Claude demonstrated different patterns of CoTR influence compared to ChatGPT. In the CoTR-disabled state (Figure 24), Claude assigned exceptionally high importance to COPD across all gender distributions (0.133-0.145), particularly for male patients, while maintaining consistent Age importance (0.149-0.163). When CoTR was enabled (Figure 25), Claude's feature importance distribution became more evenly balanced, with a noticeable decrease in the COPD-Age importance gap.

The most significant CoTR-related difference in Claude appears in ICU/intubed importance, where it is ranked significantly lower in AllFemale, AllMale, and Gender Balanced distributions when CoTR is disabled. With CoTR enabled, Claude assigned dramatically higher ICU/Intubed importance for these gender distributions when compared to the CoTR-disabled model, except for in the GenderUnbalanced distribution, which is assigned lower ICU/intubed values, suggesting that explicit reasoning pathways leads Claude to value associated features related to critical care indicators more heavily.



Figures 24, 25: Top 5 Features and Feature Importance by Gender, Claude, CoTRDisabled and CoTR Enabled

Comparative Gender Analysis Among LLMs

Our analysis of feature importance across multiple LLMs revealed consistent patterns in how these models prioritize clinical features for COVID-19 and lung disease prediction. When examining Figure 26, Age consistently appears as the most important feature across all gender distributions (0.12-0.15), followed by Intubed status (0.112) and ICU admission (0.111). Critical care features show substantial importance (0.10-0.12) across all distributions, with the AllFemalePregnant dataset showing notably higher importance for ICU and Intubed features. Gender-specific features demonstrate expected variations, with Pregnancy showing significant importance in female distributions while having near-zero importance in AllMale distribution. Comorbidities like COPD, Cardiovascular, and Diabetes show moderate importance (0.06-0.08) with subtle variations across gender distributions. These differences highlight how LLMs may incorporate demographic factors into their clinical reasoning processes, even without explicit instruction, potentially influencing predictions in ways that either mitigate or reinforce biases in medical data.

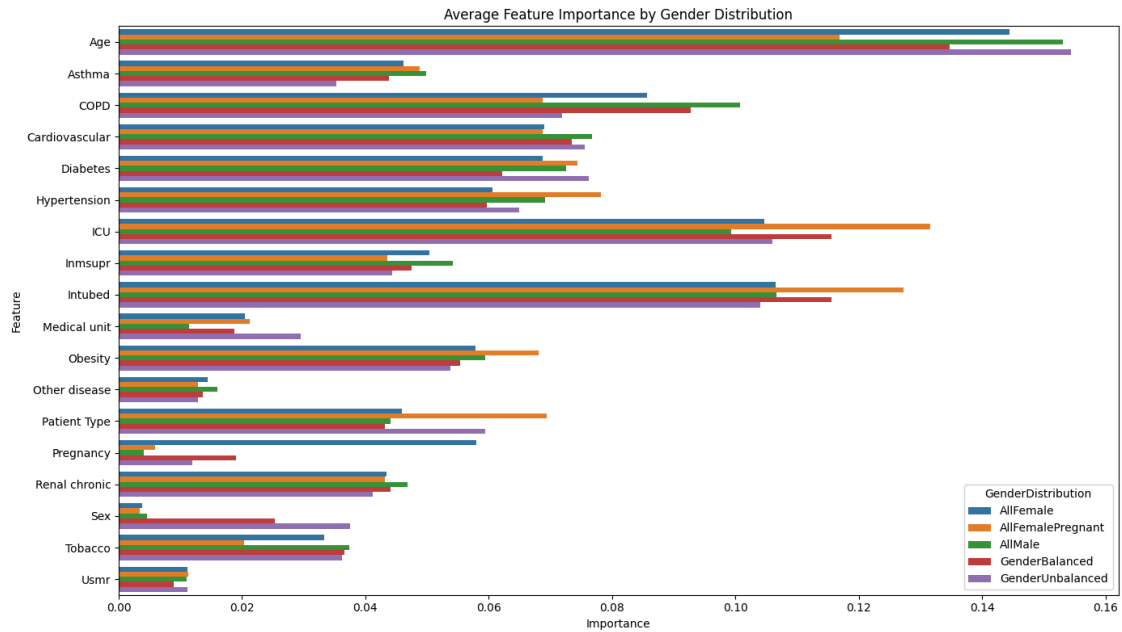


Figure 26 : Average Feature Importance by Gender Distribution

Data Prediction

For COVID-19 presence prediction, ChatGPT achieves moderate accuracy ranging from 54-65% across gender distributions, with relatively balanced performance between classes. Unlike the CML models which showed strong biases toward positive predictions, ChatGPT demonstrates more balanced recall and specificity, particularly in gender-balanced and unbalanced distributions where it achieves nearly equal true positive and true negative rates. The AllFemalePregnant distribution shows the highest accuracy (64.5%) but with significant class imbalance, as the model heavily favors negative predictions.

For COVID-19 severity prediction, ChatGPT performs poorly across all gender distributions (Figure 27), with accuracy of only 0.8-51.8%. Most strikingly, the model completely fails to predict class 0 in AllFemale and AllFemalePregnant populations, and class 3 in AllMale populations. The F1 scores are extremely low (0.006-0.22), indicating fundamental difficulties with this multiclass prediction task. This

suggests ChatGPT struggles to leverage its domain knowledge to differentiate severity levels without contextual pattern recognition that CML models can learn directly from data.

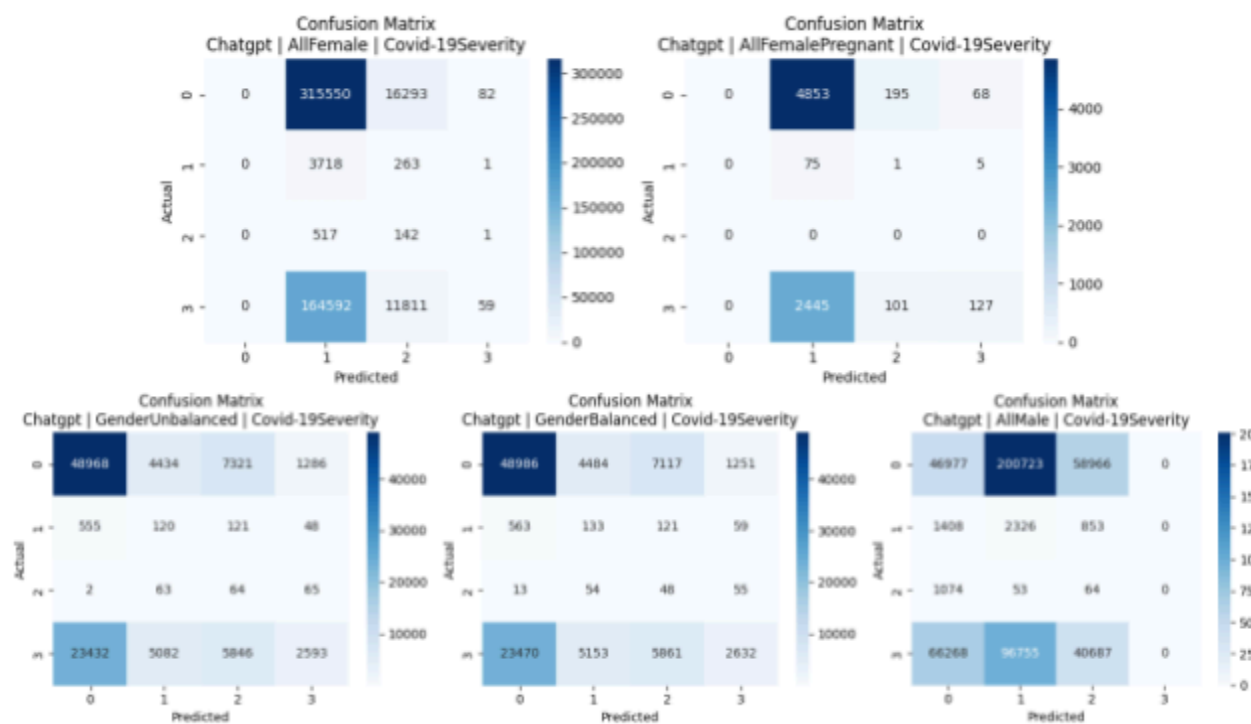


Figure 27: Confusion matrices for Covid-19 Severity Predictions, ChatGPT

For death prediction, ChatGPT achieves high accuracy (87-98.7%) but shows significantly different error patterns compared to CML models. While CML models demonstrated high recall but low specificity, ChatGPT shows the opposite pattern - high specificity but very low recall, especially in AllFemalePregnant (4.5%) and GenderBalanced (6.5%) distributions. This indicates a strong bias toward predicting non-death outcomes, potentially reflecting caution in making high-stakes predictions.

For pneumonia prediction, the model achieves good accuracy (80-89%) but similarly demonstrates high specificity and low recall, opposite to the CML models' patterns. ChatGPT correctly identifies most non-pneumonia cases but frequently misses actual pneumonia cases, with recall as low as 9.1% in the AllFemale distribution.

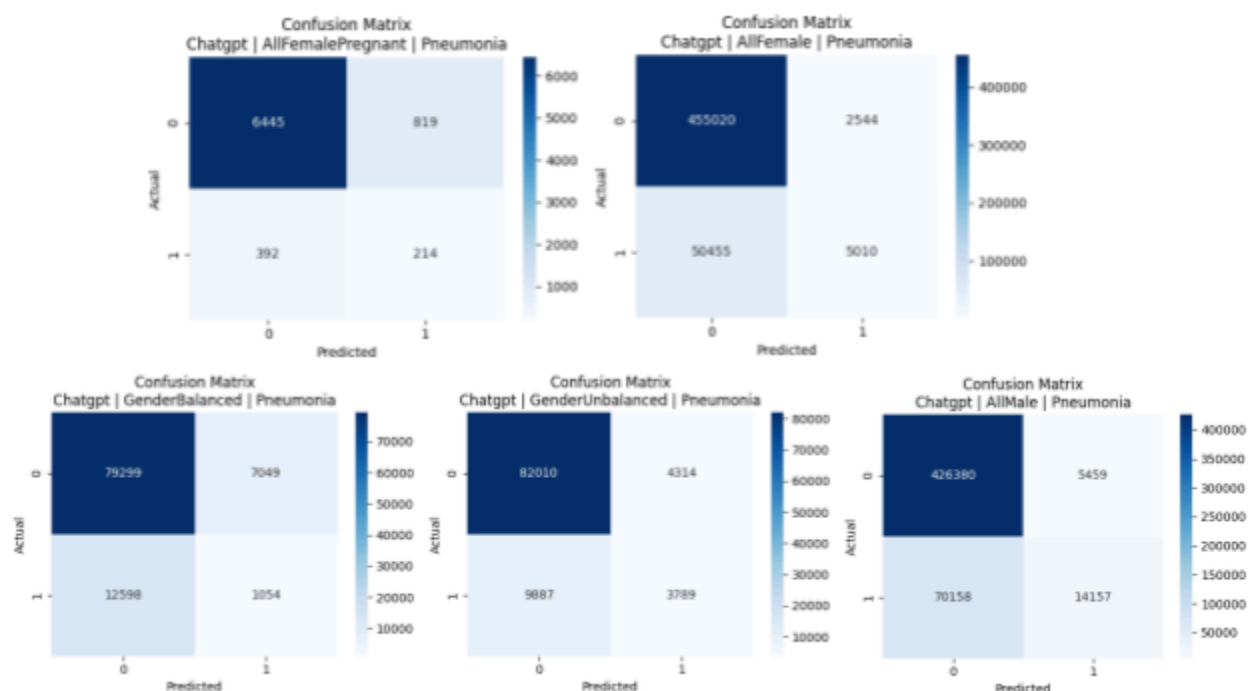


Figure 28: Confusion matrices for Pneumonia Predictions, ChatGPT

Comparison of LLM and CML for Feature Importance

The heatmap of average feature importance (Figure 29) shows striking differences in how models weight clinical features. XGBoost consistently places exceptional emphasis on age (0.32) and Catboost (0.35) compared to LLMs like Claude (0.15), Chatgpt (0.14), and Deepseek (0.18). This disparity suggests that while all models recognize age as important, CML models trained directly on patient data appear to identify stronger statistical associations with outcomes than LLMs relying on domain knowledge.

For medical unit, the difference is even more pronounced - XGBoost and Catboost assign substantial importance (0.14 and 0.13 correspondingly) while LLMs assign minimal weight (0.01 - 0.02). This highlights how CML models can identify system-level factors that may not be recognized in medical literature but emerge strongly in actual patient data.

Examining Figure 30 (Top Features from LLMs and XGBoost), we see that while both approaches agree on the significance of age, their subsequent feature rankings diverge considerably. LLMs consistently

prioritize clinical features like ICU admission (0.12) and intubation (0.12), aligning with clinical understanding of COVID-19 progression. In contrast, XGBoost places higher importance on administrative features like medical unit (0.14) and statistical demographic patterns like gender and regional markers (USMR).

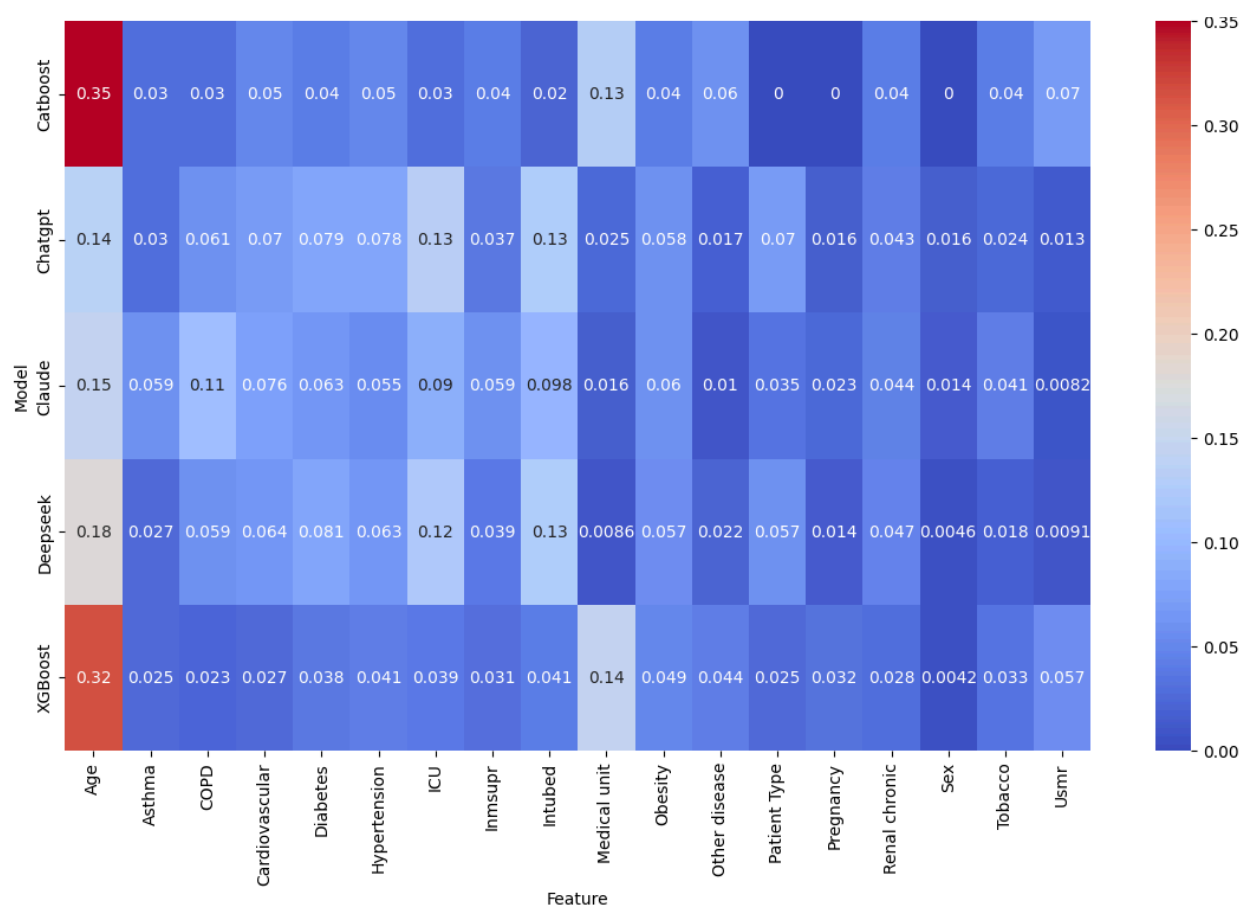


Figure 29: Average Feature Importance across LLMs and CML

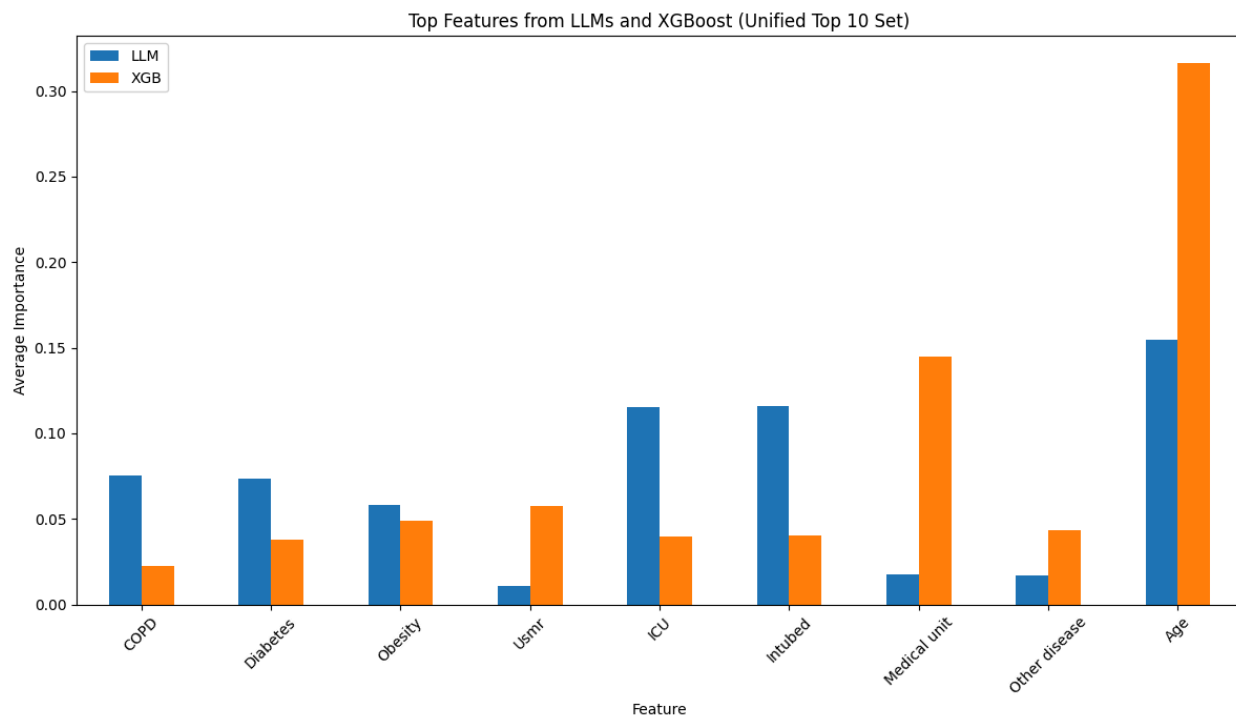


Figure 30: Top Features across LLMs and CML (XGBoost)

The bar charts in Figure 31 reveal important patterns across different gender distributions:

1. LLMs maintain more consistent feature importance rankings regardless of gender distribution, suggesting they rely on general medical knowledge rather than gender-specific statistical patterns.
2. CML models (represented by black diamonds for XGBoost and green bars for CatBoost), specifically CatBoost, show considerable variation in feature weights across gender categories. For instance, in the all-male distribution, age receives substantially higher importance than in female-only datasets for Catboost.
3. For pregnant female patients, both model types show distinct feature importance distributions, but LLMs appear more consistent in maintaining clinical priority features.

The low calculated correlation between LLM and CML feature rankings (Spearman Rank Correlation of 0.053) additionally indicates fundamentally different approaches to feature prioritization despite some agreement on top features.

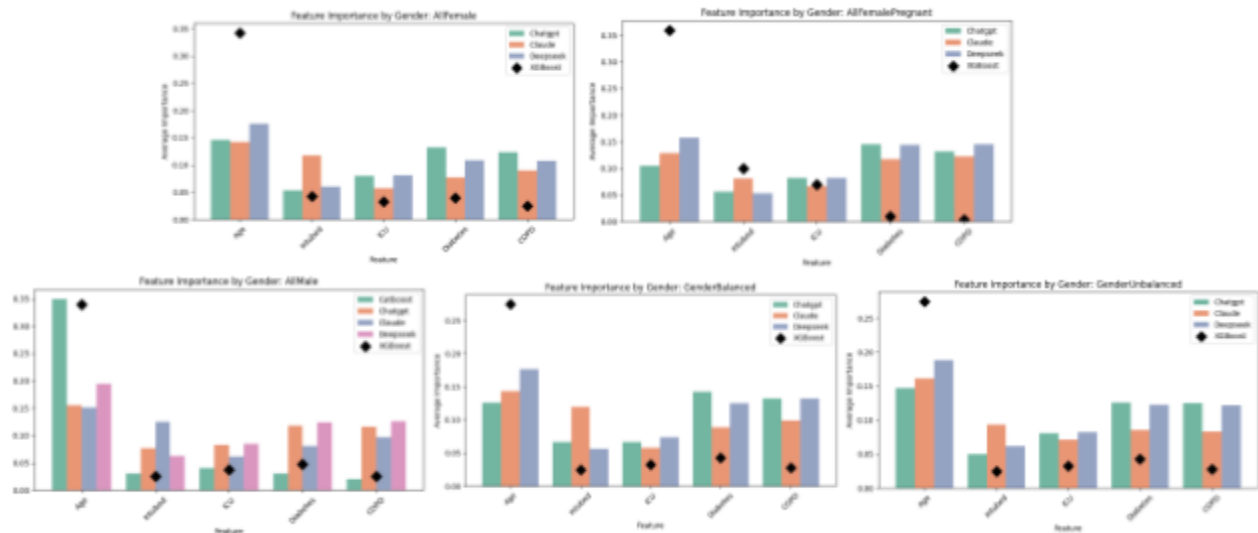


Figure 31: Top Features by Gender Distribution, CML vs. LLMs

Comparison of LLM and CML for Prediction

For COVID-19 presence prediction, XGBoost (67% accuracy) and CatBoost (65-68% accuracy) outperform ChatGPT (60% accuracy), but with different error patterns (Figure 32 and 33). CML models show high recall but low specificity (strong bias toward positive predictions), while ChatGPT demonstrates more balanced performance between classes.

COVID-19 severity prediction reveals the largest performance gap: XGBoost achieves 65-67% accuracy but only predicts extreme classes (0 and 3), ChatGPT performs poorly (8-52% accuracy) with extremely low F1 scores (0.006-0.22), and CatBoost struggles at 42-50% accuracy.

For death prediction, all models achieve high accuracy with different error patterns: XGBoost shows high recall (98-99%) but low specificity, ChatGPT demonstrates high specificity but very low recall (4.5-6.5% in some populations), and CatBoost provides the most balanced approach.

Gender-based analysis reveals XGBoost performs best on all-male populations and worst on pregnant female data, while ChatGPT and CatBoost show more consistent cross-gender performance despite lower overall accuracy.

These findings illustrate that CML models offer higher accuracy but with significant class biases, while ChatGPT exhibits more balanced predictions but lower overall accuracy, suggesting potential benefits from hybrid approaches.

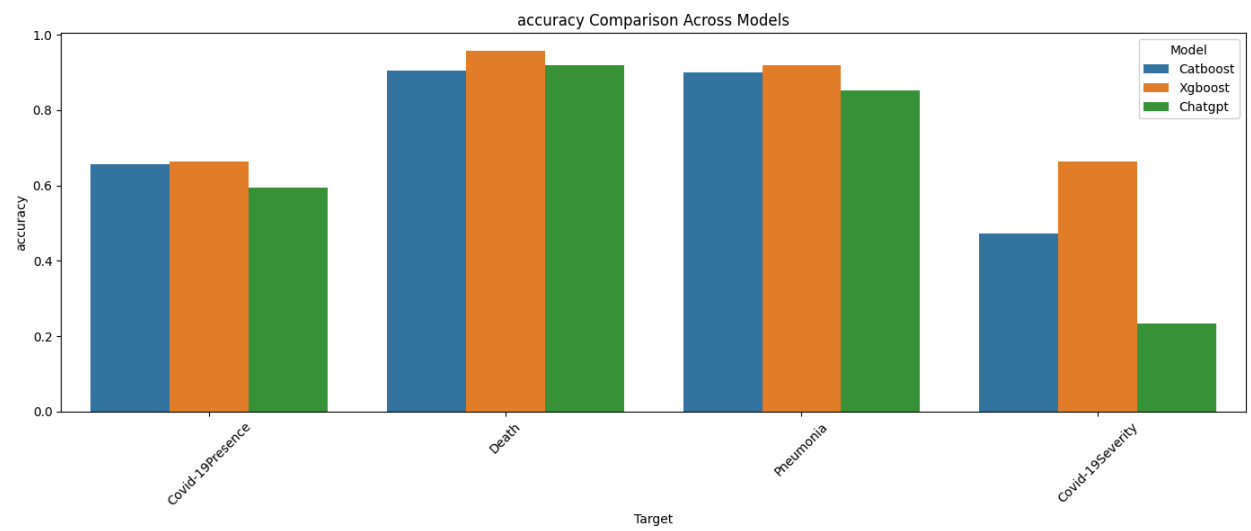


Figure 32: Prediction Accuracy Comparison across Models

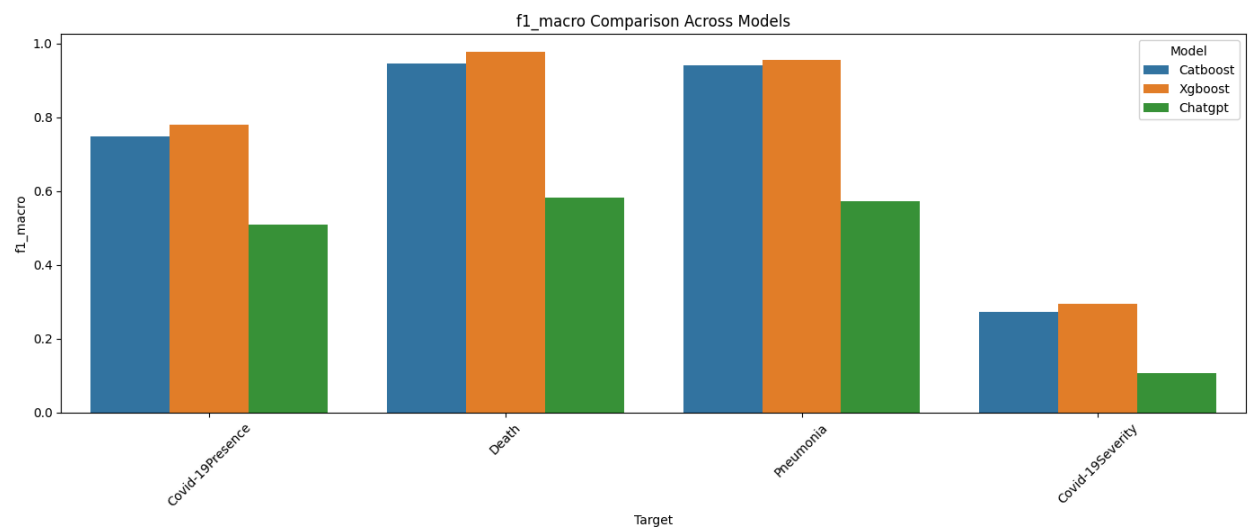


Figure 33: F1 Comparison Across Models

Deliverables

All relevant coding for LLMs/CML runs

[ML vs LLM COVID Bias Github](#)

Mexican Government dataset

[COVID-19 Dataset](#)

Skill Learning

During the course of this project, our team developed a range of meta-cognitive skills that encapsulate abilities beyond the scope of this single report

1. **Comparative Reasoning Across Model Types**

One critical skill we cultivated was the ability to evaluate the strengths and weaknesses of Classical Machine Learning (CML) versus Large Language Models (LLMs). Our experiment design made efforts to treat both fairly—through the standardization of prompt engineering, data balancing, and shared evaluation metrics—we learned fundamental differences in model outputs for head-to-head comparison. It has become incredibly essential especially now that AI technologies become more multi-modal and are used in various disciplines.

2. **Prompt Engineering and Semantic Precision**

An appreciation for the delicate art of prompting an LLM isn't just about making queries—maintaining cohesion across all models avoids ambiguity and derives interpretable outputs. From our three iterations of our feature importance prompt, we managed the subtlety of changes (e.g., requiring mathematically sound outputs, avoiding the mention of existing data) greatly impacted response quality. It sharpened our ability to create optimal prompts that are both correct and semantically robust with the domain knowledge of our models.

3. **Handling Imperfect Data**

Incorporating real-world healthcare data taught us how to assess, clean, and process datasets to account for missing values or otherwise NaN data and consider demographic skews. We kept in mind the importance of ethical challenges around gender and pregnancy representation. It helped us learn to make purposeful choices—omitting "unknown" values to ensure fairness in model comparisons and consequently reflecting a sensitivity to bias.

4. **Debugging Complex ML Pipelines**

Our encounters of major roadblocks in the form of kernel crashes and memory leaks during model training prepared us for the workload of our task. Diagnosing this required stepping through each layer of our training script to resolve tensor shape mismatches. The necessitation of manual troubleshooting strengthened our resilience in debugging unpredictable problems, a key skill in real-world machine learning deployment.

5. **Fairness-Aware Evaluations**

Fairness is not just as a model output property, but is integrated throughout our experiment's process. Designing gender-specific data splits, removing confounding features from LLM prompts, and creating adequate comparisons on feature importance across subgroups allowed the development of a new perspective of how algorithmic bias manifests—and how it might be prevented.

6. **Scientific Collaboration**

We honed our communication of the results of our study in complex workflows to provide whole context in writing. In creating consistent documentation of our metric collection process, producing structured JSON file outputs, or crafting our final report, we gained practice in research writing with an emphasis on clarity. Our writing and presentation skills were constantly tested in our team's coordination with the professor, TAs, and internal communicate to gain valuable feedback, improving our ability to collaborate on this highly technical study.

Future Work

Beyond COVID-19, this prompt-based approach for checking feature importance by gender could be used in many other areas of healthcare—like heart disease, cancer, or mental health—where men and women often show different risk patterns. It could also be useful outside of medicine, for example in credit scoring, hiring systems, or social-service referrals, wherever demographic balance matters. Trying our method in these new settings would help us see how well it holds up and reveal any biases unique to each field.

On the modeling side, bringing in open-source LLMs like LLaMA or Mistral is important because it lets us peek under the hood: unveiling hidden biases in domain knowledge which can influence how they weigh features and predict. Understanding those differences could guide practitioners toward models that are not only accurate but also more transparent and cost-effective.

Finally, looking at other demographics—race, age bands, income levels—matters because real populations are intersectional. By expanding beyond gender, we ensure our insights about feature relevance translate into fairer, more inclusive LLMs. This helps prevent scenarios where an otherwise promising model inadvertently reinforces existing biases against underrepresented or misrepresented groups.

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Appendix

Finalized Prompt Structuring for Extracting Feature Importance from LLMs

Objective: Rank the following 18 feature columns from a COVID-19 patient history dataset based on feature importance weight analysis as used with random forest/tree algorithms used by XGBoost and CatBoost. You will not be handed actual data. Base feature importance solely on the feature columns and corresponding descriptions. We are predicting *<target column>*. Output as JSON.

Task Details:

Populate the JSON file with <key, value> pairs representing: <feature column, weight distribution for feature importance>

Data Glossary:

Your dataset is *<gender distribution type>*.

- Sex: female or male.
- Age: of the patient.
- Patient Type: if the patient received in-unit care or returned home
- Pregnancy: whether the patient is pregnant or not.
- Diabetes: whether the patient has diabetes or not.
- COPD: Indicates whether the patient has Chronic obstructive pulmonary disease or not.
- Asthma: whether the patient has asthma or not.
- Inmsupr: whether the patient is immunosuppressed or not.
- Hypertension: whether the patient has hypertension or not.
- Cardiovascular: whether the patient has heart or blood vessels related disease or not.
- Renal chronic: whether the patient has chronic renal disease or not.
- Other disease: whether the patient has another disease or not.
- Obesity: whether the patient is obese or not.
- Tobacco: whether the patient is a tobacco user or not.
- Usmr: whether the patient treated medical units of the first, second or third level.
- Medical unit: type of institution of the National Health System that provided the care.
- Intubed: whether the patient was connected to the ventilator or not.
- ICU: Indicates whether the patient had been admitted to an Intensive Care Unit or not.

Output Format:

Output as a downloadable JSON file with this naming convention: **<LLM name>_<CotrEnabled>_<gender distribution type>_<target column>.json** for key value pairs <feature column, weight distribution for feature importance> where the weight distribution is on a scale [0,1] inclusive. Ensure all feature importance values sum to 1. Ensure all features are sorted from greatest to least feature importance value.

Finalized Prompt Structuring for Extracting Predictions from LLMs

Objective: Use the following data given COVID-19 patient history to predict corresponding target value. Base predicted target value solely on the feature columns and corresponding descriptions. We are predicting **Covid-19Presence**. Output as CSV; fill the missing column values for **Covid-19Presence**. Keep the naming convention the same.

Task Details:

For each data point in the CSV, fill in the **Covid-19Presence column with** predicted value given the feature values and gender distribution.

Data Glossary:

Your dataset is all male.

- Sex: female or male.
- Age: of the patient.
- Patient Type: if the patient received in-unit care or returned home
- Pregnancy: whether the patient is pregnant or not.
- Diabetes: whether the patient has diabetes or not.
- COPD: Indicates whether the patient has Chronic obstructive pulmonary disease or not.
- Asthma: whether the patient has asthma or not.
- Inmsupr: whether the patient is immunosuppressed or not.
- Hypertension: whether the patient has hypertension or not.
- Cardiovascular: whether the patient has heart or blood vessels related disease or not.
- Renal chronic: whether the patient has chronic renal disease or not.
- Other disease: whether the patient has another disease or not.
- Obesity: whether the patient is obese or not.
- Tobacco: whether the patient is a tobacco user or not.
- Usmr: whether the patient treated medical units of the first, second or third level.
- Medical unit: type of institution of the National Health System that provided the care.
- Intubed: whether the patient was connected to the ventilator or not.
- ICU: Indicates whether the patient had been admitted to an Intensive Care Unit or not.

Target Column Values:

- COVID-19Presence (1 or 2)

- 1: Positive
 - 2: Negative
- COVID-19Severity (0-3)
 - 1: Mild
 - 2: Moderate
 - 3: Severe
 - 0 : No COVID-19
- Death (1 or 2)
 - 1: Died
 - 2: Survived
- Pneumonia (1 or 2)
 - 1: Positive
 - 2: Negative

Output Format:

Output as a downloadable CSV file with the same naming convention as files uploaded:

Claude_AllMale_Covid-19Presence_LLMTargets.csv for data of *All Male type*. Ensure all target output values are within the range specified in the dictionary for target column values.

Collection of All Feature Importance Graphs for CML

https://github.com/kailen-mccauley/ML_vs_LLM_COVID_GenderBias/tree/main/Feature%20Importance%20Visualization/CML%20Feature%20Importance%20Graphs

Collection of Comparative Graphs Relative to CML/LLMs

https://github.com/kailen-mccauley/ML_vs_LLM_COVID_GenderBias/tree/main/Feature%20Importance%20Comparison/Graphs

Collection of All Feature Importance Graphs by Gender Distribution

LLM COTREnabled/Disabled:

https://github.com/kailen-mccauley/ML_vs_LLM_COVID_GenderBias/tree/14d04d7041dde50576d2a14cf8252af1ddd1f109/Feature%20Importance%20Visualization/output_gender_feature_barplots_v2

LLM vs. CML:

https://github.com/kailen-mccauley/ML_vs_LLM_COVID_GenderBias/tree/97275142c15608f23fa2428c4c81f1c10c77865c/Feature%20Importance%20Comparison/Graphs/feature_importance_by_gender